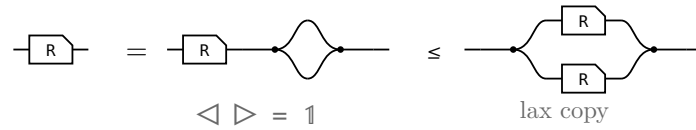


# The allegory axioms: the proofs

The companion to `allegory-axioms.typ`, which states these laws and their pictures. Here each one is worked: the steps of the Lean proof, side by side, with the rule that reaches each under it.

## §1. $R \cap R = R$ , the one that is not bookkeeping



The last picture is  $R \cap R$  itself, so this is  $R \leq R \cap R$  and the lax copy law is the whole of it. The other direction is not proved here:  $R \cap S \leq R$  holds for every  $S$ , by weakening  $S$  to  $\top$  and then collapsing  $R \cap \top$  — which is the paper’s own remaining three steps, packaged once.

## §2. The modular law, from Frobenius

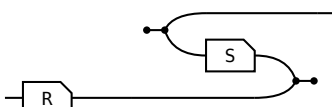
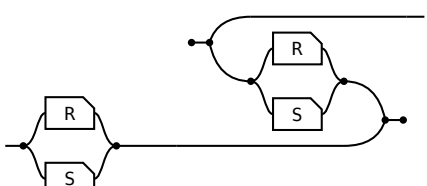
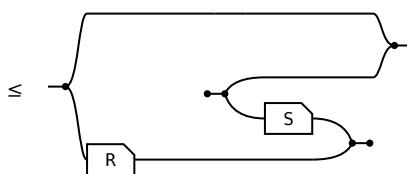
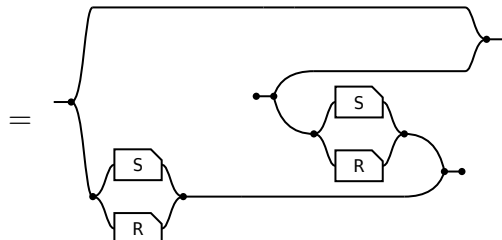
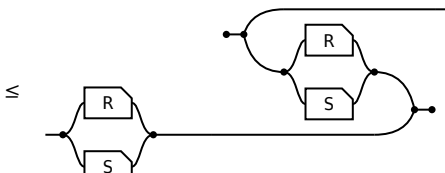
| <p>the modular law, <math>R \cap S \cap T \leq (R \cap T \cap S^\circ) \cap S</math></p> |         |                                                                                                                                 |
|------------------------------------------------------------------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------|
| term                                                                                     | picture | the rule that reaches it                                                                                                        |
| $(R \cap S) \cap T$                                                                      |         | the start: $R \cap S \cap T$ .                                                                                                  |
| $= (\triangleleft (R \otimes T)) ((S \otimes 1) \triangleright)$                         |         | Factor out the shared prefix: $(R \cap S) \otimes T$ is $(R \otimes T) (S \otimes 1)$ .                                         |
| $\leq (\triangleleft (R \otimes T)) ((1 \otimes S^\circ) (\triangleright S))$            |         | The merge slide. $S$ slides through the merge and reappears as $S^\circ$ on the other strand. The only inequality in the proof. |
| $= (R \cap (T \cap S^\circ)) \cap S$                                                     |         | Fold the prefix back in; the copy-merge pair reads as a meet again.                                                             |

$S$  occurs once on the left of the slide and twice on the right, and the lax copy law is the only law in the definition that may duplicate a box — so that is where the inequality has to be.

| <p>the merge slide, <math>(S \otimes 1) \triangleright \leq (1 \otimes S^\circ) \triangleright S</math></p> |         |                                             |
|-------------------------------------------------------------------------------------------------------------|---------|---------------------------------------------|
| term                                                                                                        | picture | the rule that reaches it                    |
| $(S \otimes 1) \triangleright$                                                                              |         | the start: $(S \otimes 1) \triangleright$ . |

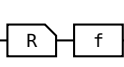
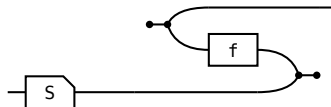
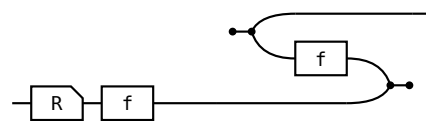
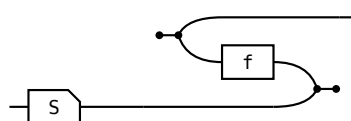


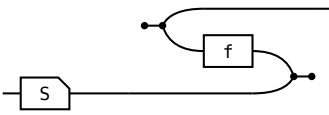
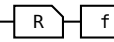
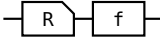
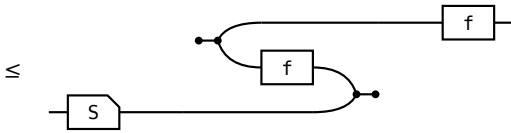
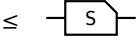


| $\Leftarrow$ given $\text{---} \leq$  show $R \cap S$ entire, $\text{---} \leq$  |                                                                                             |                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| term                                                                                                                                                                                                                                                 | picture                                                                                     | the rule that reaches it                                                                                                                                                                                                                                                                                    |
| 1                                                                                                                                                                                                                                                    | $\text{---}$                                                                                | the start: 1.                                                                                                                                                                                                                                                                                               |
| $\leq 1 \cap (R S^\circ)$                                                                                                                                                                                                                            | $\leq$    | the hypothesis, met with 1.                                                                                                                                                                                                                                                                                 |
| $= 1 \cap ((S \cap R) (S \cap R)^\circ)$                                                                                                                                                                                                             | $=$       | $1 \cap R S^\circ = 1 \cap (S \cap R) (S \cap R)^\circ$ . $R S^\circ$ relates $a$ to $a'$ when some $b$ has $a R b$ and $a' S b$ ; the $1 \cap$ sets $a' = a$ , and it reads “some $b$ has $a R b$ and $a S b$ ” — one arrow of $R \cap S$ . Two arrows become one, and that is what needs the modular law. |
| $\leq (R \cap S) (R \cap S)^\circ$                                                                                                                                                                                                                   | $\leq$  | $X \cap Y \leq Y$ , then $S \cap R = R \cap S$ .                                                                                                                                                                                                                                                            |

## §6. The shunting rule

A map moves from one side of a containment to the other at the cost of its converse,  $R f \leq S \iff R \leq S f^\circ$ . It is how a function gets out of the way so the relation underneath can be reasoned about. Each direction spends exactly one half of **map** — never both.

| <b>shunting right, <math>\Rightarrow</math></b> given $\text{---} \text{---} \leq \text{---}$  show $\text{---} \text{---} \leq \text{---}$  |                                                                                             |                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| term                                                                                                                                                                                                                                                                                                                 | picture                                                                                     | the rule that reaches it                                                   |
| R                                                                                                                                                                                                                                                                                                                    |          | the start: R.                                                              |
| $\leq R (f f^\circ)$                                                                                                                                                                                                                                                                                                 | $\leq$  | <b>total:</b> $1 \leq f f^\circ$ , so $f f^\circ$ may be inserted after R. |
| $\leq S f^\circ$                                                                                                                                                                                                                                                                                                     | $\leq$  | Re-bracket to $(R f) f^\circ$ and apply the hypothesis.                    |

|                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                          |                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>shunting right</b> , $\Leftarrow$ given  $\leq$  show  $\leq$  |                                                                                          |                                                                   |
| <b>term</b>                                                                                                                                                                                                                                                                                                                                                                                                | <b>picture</b>                                                                           | <b>the rule that reaches it</b>                                   |
| $R \ f$                                                                                                                                                                                                                                                                                                                                                                                                    |         | <b>the start:</b> $R \ f$ .                                       |
| $\leq S \ (f^\circ \ f)$                                                                                                                                                                                                                                                                                                                                                                                   |        | Apply the hypothesis on the left; the two fs are now adjacent.    |
| $\leq S$                                                                                                                                                                                                                                                                                                                                                                                                   | $\leq$  | <b>single valued:</b> $f^\circ \ f \leq 1$ , so the pair cancels. |

Shunting left is the mirror, same shape, map on the other side throughout. What the rule is, is an adjunction:  $f \dashv f^\circ$ , with **total** the unit and **single valued** the counit, so “f is a map” and “f has a right adjoint, namely  $f^\circ$ ” are one statement.